

Modeling laryngeal biomechanical control of acoustic parameters using machine learning regressors

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Accurate modeling of laryngeal motor control is key to understanding typical and disordered voice production. Traditional speech motor control models often overlook the role of intrinsic laryngeal muscles, such as the cricothyroid (CT) and thyroarytenoid (TA) muscles, and subglottal pressure (P_S) in shaping vocal output. While the recent LaDIVA model addresses this limitation via a lookup table, its limited flexibility, high computational demands, and absence of individualized vocal fold (VF) representations make comprehensive exploration of diverse motor patterns challenging. Thus, this study evaluates efficient feature-driven machine learning regressors as forward models for laryngeal control, mapping motor inputs (P_S and muscle activation levels of CT and TA) to acoustic outputs (fundamental frequency and sound pressure level). Training data were generated using two biomechanical VF models, i.e., the extended body-cover model and the triangular body-cover model. Methods using random forest (RF), neural network (NN), and polynomial regressor (PR) approximations were compared in terms of prediction accuracy, computational efficiency, and robustness to dataset size. An inverse Jacobian-based feedback controller was implemented to assess the ability of the regressors to track acoustic glides. Results show that RF achieved the highest accuracy at the expense of increased computational cost, while PR and NN offered faster execution and smaller model sizes with competitive tracking performance. These findings support incorporating ML-based forward models into laryngeal control frameworks, providing physiologically grounded and computationally efficient alternatives to the lookup table approach used in the original LaDIVA model.

I. INTRODUCTION

Biomechanical control of the larynx is essential for regulating the motion of the vocal folds (VFs) and ensuring precise phonation. This intricate control mechanism enables variations in pitch, loudness, voice quality, and phonatory stability, which are essential for intelligible speech (Chhetri and Neubauer, 2015; Zhang, 2016). Two key acoustic output parameters, fundamental frequency (f_o) and sound pressure level (SPL), are governed by the coordinated action of subglottal pressure (P_S) and intrinsic laryngeal muscle (ILM) activations, particularly the cricothyroid (a_{CT}) and thyroarytenoid (a_{TA}) (Chhetri and Park, 2016). Despite the apparent simplicity of these outputs, a deeper understanding and accurate modeling of their interactions is necessary to advance speech motor control theories (Guenther, 2016) and to investigate how biomechanical alterations in laryngeal control contribute to voice disorders (Gomez *et al.*, 2007; Hillman *et al.*, 2020).

Traditional speech motor control models, such as Directions into Velocities of Articulators (DIVA) (Tourville and Guenther, 2011) and the Feedback-Aware Control of Tasks in Speech (FACTS) model (Parrell *et al.*, 2019), have significantly advanced our understanding of speech acquisition, planning, and execution, focusing primarily on articulatory control and sensorimotor adaptation. However, these models focus their efforts on the vocal tract and treat the glottal source as a fixed waveform rather than an actively controlled element of voice production, limiting their ability to capture the physiology and the dynamic role of laryngeal biomechanics in phonation. For instance, DIVA employs a simplified glottal wave

model (Fant *et al.*, 1985) along with the Maeda vocal tract (Maeda, 1982), while FACTS uses a similar source-filter approach within the CASY synthesizer (Iskarous *et al.*, 2003).

To overcome this limitation, recent advances have incorporated laryngeal biomechanical control into speech motor control frameworks, improving physiological relevance (Palaparthi *et al.*, 2024; Weerathunge *et al.*, 2022). For example, the Laryngeal DIVA (LaDIVA) model (Weerathunge *et al.*, 2022) extends DIVA by integrating ILM activation and P_S , using the self-sustained extended body-cover model (ext-BCM) (Zañartu *et al.*, 2014) to map laryngeal control parameters, referred to as VF mobility space trajectories, into auditory task space trajectories (f_o and SPL). This approach provides a dynamic representation of VF motion, closely aligning with the mechanisms of real speech motor control and emphasizing the importance of laryngeal biomechanical modeling.

Beyond its role in speech motor control, biomechanical modeling of laryngeal dynamics is crucial for understanding the mechanisms underlying voice disorders. Voice disorders such as VF paralysis, dysphonia, and phonotrauma disrupt laryngeal biomechanics, altering VF vibration and leading to severe voice quality degradation (Chung *et al.*, 2022; Erath *et al.*, 2011; Van Houtte *et al.*, 2011). Although clinical diagnostics, including acoustic analysis, perceptual evaluation, and imaging techniques, provide useful insights, they lack the ability to assess the interconnected biomechanical mechanisms underlying phonatory deficits. These limitations motivate the development of computational models capable of capturing VF dynamics and neuromuscular interactions.

Modeling laryngeal biomechanical control is challenging due to the nonlinear and highly dynamic behavior of the VFs. A range of numerical models has been developed to investigate

different aspects of phonation, each balancing computational efficiency and physiological realism. Lumped-element models, such as the Body Cover Model (BCM) and Triangular Body Cover Model (TBCM) (Alzamendi *et al.*, 2022; Galindo *et al.*, 2017; Story and Titze, 1995; Zañartu *et al.*, 2014), use mass-spring-damper representations to approximate VF mechanics. These models provide a reasonable compromise between computational cost and accuracy (Erath *et al.*, 2013). In contrast, three-dimensional finite element models (Hunter *et al.*, 2004; Zhang, 2017) simulate VF tissue deformation and aerodynamics in greater detail but require substantially higher computational resources.

Despite their balance of efficiency and physiological accuracy, lumped-element models are difficult to apply directly in real-time control schemes, as iterative numerical solutions for direct and inverse solutions at each time step are challenging. Precomputed phonatory outcomes from physics-based simulations offer a solution, reducing the need for continuous differential equation solving while retaining physiological relevance. For example, LaDIVA uses a large lookup table (LUT) to store precomputed ext-BCM simulation features, allowing laryngeal control parameters to be mapped to acoustic outputs in real time (Weerathunge *et al.*, 2022). However, LUT-based models cannot readily adapt to dynamic phonatory conditions, demand substantial memory, lack the flexibility for subject-specific fine-tuning, and are prone to interpolation errors when values fall outside the precomputed parameter space, particularly in highly nonlinear phonatory dynamics.

These challenges motivate the use of data-driven approaches such as machine learning (ML), which can approximate nonlinear mappings while remaining adaptable to individual variability. In voice production, ML-based models have shown promise in estimating laryn-

geal source parameters from acoustic and aerodynamic features, including P_S (Gomez *et al.*, 2007; Lin *et al.*, 2020; Marks *et al.*, 2020), VF parameters (Donhauser *et al.*, 2024; Zhang, 2020), and muscle activation (Ibarra *et al.*, 2021; Sepúlveda *et al.*, 2024). Furthermore, ML approaches enable subject-specific customization through transfer learning, allowing general models to be easily adapted to individual laryngeal mechanics (Ibarra *et al.*, 2025), while also expanding the training set with Bayesian methods (Sepúlveda *et al.*, 2024). This combination of robustness and adaptability enhances modeling precision, making ML-based approaches particularly valuable for personalized voice assessment and clinical applications.

ML techniques for estimating laryngeal biomechanical parameters are primarily based on feature-driven regressors or direct signal-based learning. Feature-driven models extract meaningful biomechanical representations from data, offering efficient estimation of laryngeal control parameters with greater interpretability and lower computational cost (Ibarra *et al.*, 2021; Zhang, 2020). In contrast, deep learning approaches, such as recurrent neural networks (RNNs) (Gómez *et al.*, 2019) and convolutional recurrent neural networks (CRNNs) (Donhauser *et al.*, 2024), rely on raw signal inputs like VF trajectory time series. These signal-based models are computationally expensive, require large datasets that are often unavailable for clinical use, and lack interpretability. Since capturing the complex nonlinear interactions of laryngeal biomechanics is critical, feature-driven regressors represent a more viable option for biomechanical laryngeal motor control, offering a balance between computational efficiency and physiological accuracy for integration into control frameworks.

In this study, the accuracy and generalizability of feature-driven ML regressors for estimating laryngeal biomechanical parameters from acoustic outputs were evaluated. Two

physiologically grounded lumped-element models, the ext-BCM (Zañartu *et al.*, 2014) and TBCM (Alzamendi *et al.*, 2022), were used to simulate VF dynamics under a range of laryngeal control inputs. Using these simulations as training data, multiple ML algorithms, including gradient-boosted decision trees and random forest regressors, were trained to predict ILM activation and P_S from simulated acoustic and aerodynamic features. By evaluating ML performance with physiologically accurate VF simulations, we aim to provide a pathway toward real-time, subject-specific control in LaDIVA and related frameworks.

This study evaluates feature-driven ML regressors within laryngeal biomechanical control. As the forward model component, the regressors map input parameters (P_S , a_{CT} , a_{TA}) to acoustic outputs (f_o and SPL), and their performance is assessed in terms of accuracy, model size, execution time, and retraining capability across different VF models. Two lumped-element VF models are used, i.e., the ext-BCM, implemented as a LUT in LaDIVA (Weerathunge *et al.*, 2022), and the TBCM (Alzamendi *et al.*, 2022), which extends physiological complexity by incorporating five ILM activations and vocal posture dynamics. As part of the feedback controller, an Inverse Jacobian method iteratively adjusts input parameters by estimating how variations influence acoustic outputs. This stage is evaluated based on trajectory tracking performance, determining which regressor ensures the most precise and adaptive phonatory control for each VF model.

By systematically comparing ML regressors, this study advances laryngeal biomechanical modeling in multiple ways. First, it identifies the most effective ML strategy for laryngeal control based on accuracy, computational efficiency, and adaptability. Second, the integration of ML regressors into a two-stage control framework improves the precision and flexi-

bility of phonatory simulations relative to conventional LUT-based approaches. Finally, the comparative evaluation across two VF models provides insights into the generalizability of ML techniques across biomechanical contexts. These results contribute to the development of efficient, physiologically grounded ML models, supporting both fundamental research in laryngeal biomechanics and potential clinical or assistive applications.

II. MATERIALS AND METHODS

Speech motor control models comprise three main components: the planner, the controller, and the plant (Guenther, 2016; Parrell *et al.*, 2019). The planner encodes desired sensory targets, including acoustic and somatosensory goals. The controller integrates feed-forward control, generating motor commands based on learned relationships between physiological inputs and sensory feedback, and feedback control, which monitors deviations from intended speech via sensory feedback and guides corrective actions. The plant represents the physical system, including the articulatory vocal tract, vocal folds, and somatosensory components. A schematic of these components is shown in Figure 1.

This study focuses on modeling laryngeal biomechanical control of acoustic components. No vocal tract or somatosensory model is considered for simplicity. The control target is defined by two key acoustic trajectories, specifically pitch (f_o) and loudness (SPL) glides. The plant is represented by different ML regressors, which approximate the behavior of numerical lumped-element VF models (ext-BCM or TBCM) and provide a data-driven representation of their dynamic response. The plant functions as a forward mapping from motor inputs (P_S , a_{CT} , and a_{TA}) to acoustic outputs (f_o and SPL), enabling feedforward predictive control.

The controller solves an inverse problem using an inverse Jacobian approach (Weerathunge *et al.*, 2022) together with the forward mapping, allowing adaptive adjustments to minimize deviations from the intended acoustic trajectories. By integrating ML-based approximations of ext-BCM or TBCM within a forward-feedback control framework, this initial approach aims to capture the complex interplay between physiological inputs and acoustic outputs, providing a coherent model of laryngeal biomechanical control over the selected acoustic components.

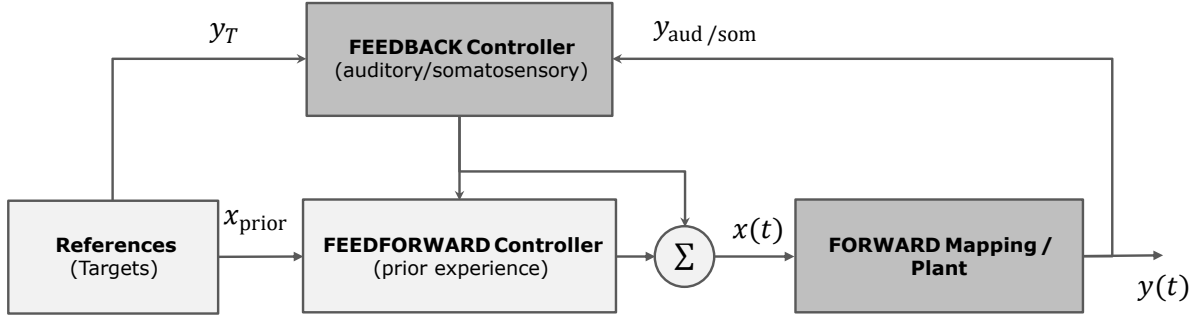


FIG. 1: Schematic representation of the biomechanical laryngeal control model.

The methodology involves training and evaluating various ML regressors to determine the most effective model for the forward mapping. To achieve this, various ML regressors are trained using synthetic data generated from lumped-element VF models, and their performance is evaluated based on prediction accuracy, computational efficiency, and generalization to different laryngeal configurations. To incorporate the feedback component into this evaluation, dynamic control exercises are designed. The goal is to analyze how the selected ML regressor influences the correction process, particularly in minimizing deviations between the estimated and target dynamic trajectories, while also considering the dependence on the biomechanical model or the dataset used for training.

A. Vocal fold models and dataset generation

Lumped-element biomechanical models provide computationally efficient representations of the dynamics of VF by approximating their motion using mass-spring-damper systems. These models capture key biomechanical interactions while maintaining a balance between physiological accuracy and computational feasibility. In this study, we selected two lumped-element VF models, ext-BCM and TBCM, based on their ability to capture different aspects of phonation dynamics: the ext-BCM provides a simplified yet physiologically grounded representation of vocal fold motion, while the TBCM extends this framework by incorporating a more anatomically detailed triangular configuration. These models were used to generate synthetic datasets for training ML regressors, which serve as the computational plant in the proposed control framework.

To construct the training datasets, simulations were performed under sustained vowel phonation conditions, systematically varying the motor inputs, P_S , a_{CT} , a_{TA} , in physiologically relevant ranges. For each simulation, acoustic outputs (f_o and SPL) were extracted from a stable 50 ms portion of the phonatory signal to ensure consistency in feature representation. Specifically, f_o was derived from the simulated glottal airflow waveform, while SPL was obtained from the radiated sound pressure signal in front of the lips with no time or frequency weighting.

The ext-BCM dataset was generated using the VF model described in [Zañartu et al. \(2014\)](#), which is implemented as a plant to be controlled in the LaDIVA model through an LUT ([Weerathunge et al., 2022](#)). This VF model extends the classical BCM originally

proposed by [Story and Titze \(1995\)](#), differentiating between the body and cover layers of the vocal folds while incorporating a fixed posterior glottal opening. In addition, it models neuromuscular activation of key ILM, cricothyroid (CT), thyroarytenoid (TA), and lateral cricoarytenoid (LCA), along with P_S . These activations influence VF stiffness and geometry ([Titze and Story, 2002](#)), with muscle activation levels normalized between 0 (rest) and 1 (maximum contraction). For dataset generation, LCA activation was fixed at 0.5, while a_{CT} and a_{TA} were varied from 0 to 1 in 0.01 increments, and P_S was adjusted from 10 to 2010 Pa in 25 Pa increments. This resulted in a comprehensive dataset consisting of about 800,000 samples.

The TBCM dataset was generated using the TBCM model described in ([Alzamendi et al., 2022](#)), which extends the BCM framework by incorporating a triangular VF configuration ([Galindo et al., 2017](#)) to more accurately represent glottal dynamics during phonation. This model was selected because its triangular representation accounts for spatial variations in vocal fold motion, providing a more detailed characterization of phonatory biomechanics. It includes five ILM—TA, CT, LCA, interarytenoid (IA), and posterior cricoarytenoid (PCA)—which interact to regulate VF stiffness, mass distribution, and vocal posture ([Titze and Hunter, 2007](#); [Titze and Story, 2002](#)). The dataset simulates typical phonation by systematically varying a_{CT} and a_{TA} from 0 to 1 in 0.025 increments and P_S from 500 to 2000 Pa in 50 Pa increments. To ensure complete glottal closure, LCA and IA activations were fixed at 0.5, while PCA activation was set to 0.1. All simulations were conducted using a male phonatory system configuration, yielding approximately 50,000 samples.

B. Machine learning regressors

For the forward mapping component, we assess the performance of five machine learning regressors in replicating the forward transformation derived from ext-BCM simulations. These regressors are categorized into three main types based on their learning mechanisms and their ability to model nonlinear relationships:

Tree-Based Methods: Random Forest (RF), Gradient Boosting (GB) and Extreme Gradient Boosting (XGB) employ decision tree structures and ensemble learning techniques to capture complex, nonlinear dependencies.

Neural Network-Based Regression: A neural network (NN) regressor was implemented as a multilayer perceptron (MLP), designed to model nonlinear relationships between input and output variables. Polynomial Regression: The Polynomial Regressor (PR) extends linear regression by fitting polynomial functions to input variables, allowing for nonlinear approximations.

The tree-based regressors were implemented using the Scikit-learn library with the “MultiOutputRegressor” strategy, which allows fitting a separate model for each target variable in parallel. The NN regressor consists of an input layer with three neurons, three hidden layers with 128 neurons each, and an output layer with two neurons, inspired in previous regression work ([Ibarra *et al.*, 2021](#)). Each hidden layer employs a Rectified Linear Unit (ReLU) activation function and incorporates a dropout layer of 0.1 to mitigate overfitting. The PR was implemented using the “PolynomialFeatures” function in Scikit-learn, which generates

a new feature matrix consisting of all polynomial combinations of the input features up to a specified degree.

For all regressors, the inputs were a_{CT} , a_{TA} , and P_S , while the outputs were f_o and SPL. The dataset used for training was derived from the ext-BCM model and underwent min-max normalization. The training procedure was organized in two stages. In the first stage, the dataset was randomly split into 80% training and 20% testing to perform an initial evaluation of the three tree-based regressors (RF, GB, XGB) using their default configurations. This step allowed the selection of the most promising tree-based method to be studied in detail. In the second stage, hyperparameter tuning was conducted for the selected tree-based model, the PR, and the NN. This process relied on 5-fold cross-validation applied to the entire dataset, ensuring that every data subset was used for both training and testing in a rotational manner. Importantly, the validation folds acted as independent test subsets that did not participate in model fitting, providing unbiased estimates of generalization performance.

For the PR, the degree was varied from 1 to 15. For the tree-based model, the number of estimators was varied between 50 and 200, and the maximum depth between 20 and 60. Hyperparameter tuning was performed using the "GridSearchCV" method from Scikit-learn, employing 5-fold cross-validation to identify the optimal configuration for each regressor (Pedregosa *et al.*, 2011). For the NN regressor, the same 5-fold strategy was applied: the validation fold in each split was used exclusively for performance evaluation, without being incorporated into gradient descent optimization or early stopping criteria.

Model performance was assessed using mean absolute error (MAE), root mean square error (RMSE), and the coefficient of determination (R^2). For each regressor, the optimal

configuration was selected based on the best-performing fold, reflecting the model that most accurately captured the input–output mapping.

All training and evaluation procedures were executed on a desktop computer equipped with an Intel i5-10400F processor, 128 GB of RAM, and an NVIDIA RTX 3050 GPU.

Finally, the best-performing ML regressors trained on the ext-BCM dataset were re-trained from scratch using the TBCM dataset to evaluate its adaptability across different laryngeal representations. Keeping the same data split strategy, the model was evaluated using MAE, RMSE, and R^2 . This step was included to test the capacity of the regressors to generalize across datasets of different magnitudes, while the use of synthetic data in the first stage provides a robust foundation for subsequent fine-tuning with more limited experimental datasets.

C. Feedback error using inverse Jacobian method for reference tracking

Following LaDIVA framework (Weerathunge *et al.*, 2022), the general feedback controller transforms the differences between the target output and the estimated auditory/somatosensory output into corrections in the motor command space. The feedback mechanism utilizes a linear approximation of the forward mapping $F\{x\}$ around a point x_0 is given by:

$$F\{x\} = F\{x_0\} + \mathbf{J}(x_0)(x - x_0) + O(2), \quad (1)$$

where $\mathbf{J}$ is the Jacobian matrix, composed of all partial derivatives $J_{i,j}(x_0) = \frac{\partial F_j}{\partial x_i} \Big|_{x_0}$, and $O(2)$ represents the exclusion of quadratic and higher-order terms, which are considered negligible. By isolating the linear component, this equation establishes a connection between

272 differences in the output space and those in the input space, leading to the update rule for
 273 x_0 to transition towards x :

$$x = x_0 + \mathbf{J}^{-1}(x_0)(F\{x\} - F\{x_0\}). \quad (2)$$

274 The inverse Jacobian, $\mathbf{J}^{-1}$, is computed using a modified Moore-Penrose pseudoinverse
 275 incorporating a damped least squares constraint (Deo and Walker, 1995). This technique
 276 introduces a penalization term in the inverse Jacobian calculation, ensuring that the solu-
 277 tion for x remains close to the initial condition x_0 , thus improving numerical stability and
 278 robustness.

$$\mathbf{J}^{-1}(x, t) = \mathbf{J}^T(x, t) (\mathbf{J}(x, t)\mathbf{J}^T(x, t) - \gamma^2\mathbf{I})^{-1}, \quad (3)$$

279 where, for $\gamma \rightarrow 0$, $\mathbf{J}^{-1}$ reduces to the classical Moore-Penrose pseudoinverse of $\mathbf{J}$. An ad-
 280 ditional constraint for the inverse Jacobian on the input space can be introduced. Prior
 281 studies in robotics have explored methods for achieving smooth control in redundant sys-
 282 tems (Guenther and Barreca, 1997; Nakamura *et al.*, 1987), employing the the null-space
 283 projection of $\mathbf{J}$ to minimize a function $H(x)$, typically the distance between the input and
 284 a reference point in the motor command space ($H(x) = \|x - x_c\|$), where x_c represents this
 285 reference. This constraint adds an extra term to the feedback expression:

$$x' = (\mathbf{J}^{-1}(x, t)\mathbf{J}(x, t) - \mathbf{I}) \nabla H(x), \quad (4)$$

286 where $\mathbf{J}^{-1}$ is computed from Eq. 3.

287 With these two elements, the feedback error is defined as:

$$error_{\text{Jac}} = \mathbf{J}^{-1}(x, t)(y_T(t) - y(t)) - \beta (\mathbf{J}^{-1}(x, t)\mathbf{J}(x, t) - \mathbf{I}) \nabla H(x), \quad (5)$$

where y_T and y denote the target output and the current output, respectively, and β is the gain for the null-space constraint. The update equation for the input space is then given by:

$$x(t + \Delta t) = x(t) + \alpha \text{error}_{\text{Jac}} \quad (6)$$

Its important to note that equations 3, 5 and 6 contain adjustable parameters (α, β, γ). For gesture tracking, x is the input vector (a_{CT}, a_{TA}, P_S) and F is the output vector (f_o , SPL). The Jacobian parameters were set to $\alpha = 0.1$, $\gamma = 0.1$, $\beta = 0$, which means that no contribution of null space was considered. The numerical derivative was evaluated using a step size of 0.02 for the normalized input.

This feedback component was used in the control experiment for the models trained with ext-BCM and TBCM datasets. The tracking gestures followed a glide pattern, specifically a pitch glide, characterized by a quadratic increase in f_o from 150 Hz to 300 Hz while maintaining a constant SPL of 70 dB, and a loudness glide, in which the SPL increased linearly from 65 dB to 85 dB for ext-BCM and from 75 dB to 110 dB for TBCM, while maintaining a constant f_o of 200 Hz.

This control or tracking exercise aims to evaluate the effectiveness of ML-based regressors in generating smooth, continuous estimates for trajectory tracking. Additionally, it assesses whether the model performance is influenced by training on datasets with different orders of magnitude in the number of examples.

III. RESULTS

This section presents the evaluation of ML regressors for forward mapping and reference tracking using both the ext-BCM and TBCM datasets. The results for each dataset are structured as follows. First, the performance of tree-based regressors is evaluated; next, optimized regressors from different model families are assessed. Then, the predictions of each optimized ML regressor are compared to the original dataset to evaluate accuracy. Finally, the ability of each optimized regressor to track reference trajectories for pitch and loudness glide gestures within a control scheme is analyzed.

A. Evaluation of machine learning regressions for forward mapping and reference tracking scheme using the ext-BCM Dataset

The initial evaluation was performed using the ext-BCM dataset, which served as the primary training set for the regressors. The first step was to select the best tree-based regressor. The results for this family of regressors are presented in Table I, comparing the MAE, RMSE and R^2 scores of the estimated f_o and SPL. The RF and XGB regressors achieved the highest R^2 scores of 0.99. Among the other metrics, RF stood out by presenting the lowest estimation errors, with MAE and RMSE below 1 Hz for f_o and below 0.1 dB for SPL. Therefore, RF was selected as the best tree-based regressor for further optimization.

Table II summarizes the performance of the optimized regressors, including models from other families (NN and PR). All three regressors achieved excellent R^2 scores, exceeding 0.998 in all cases. The optimized RF regressor exhibited the lowest estimation errors, with

TABLE I: Error metrics for initial direct mapping fitting of ext-BCM f_o and SPL using tree-based methods. RF: Random Forest, GB: Gradient Boosting, XGB: Extreme Gradient Boosting.

	f_o			SPL		
	R^2	MAE (Hz)	RMSE (Hz)	R^2	MAE (dB)	RMSE (dB)
RF	0.999	0.355	0.578	0.999	0.053	0.074
GB	0.985	8.907	12.656	0.984	0.845	1.033
XGB	0.999	1.368	1.854	0.999	0.084	0.120

less than 0.5 Hz for f_o and less than 0.05 dB for SPL. In terms of f_o , the NN and RF regressors had similar RMSE values, approximately 1.5 Hz, while for the remaining metrics, PR showed a slight improvement over NN.

To further compare these regressors, Figure 2 illustrates the computational time as a function of sample size. The results represent the mean execution time over 10 runs for different sample sizes, displayed on a logarithmic scale for better visualization of variations at smaller sizes. The PR regressor was the fastest, being half an order of magnitude faster than RF and one order of magnitude faster than NN. Specifically, PR required 7×10^{-4} s for a single sample (a 3-component vector) and 4 s for a dataset of size 10^6 . On the other hand, NN initially required the longest time to generate an estimate (5×10^{-2} s); however, for sample sizes above 1k examples, it overtook RF in efficiency. At 10^6 samples, NN required

TABLE II: Error metrics after cross-validation and hyper-parameter fitting for direct mapping fitting of ext-BCM f_o and SPL using Random Forest (RF), Neural Network (NN) and Polynomial Regressor (PR).

	f_o			SPL		
	R^2	MAE (Hz)	RMSE (Hz)	R^2	MAE (dB)	RMSE (dB)
RF	0.999	0.172	0.473	0.999	0.023	0.039
NN	0.999	0.878	1.491	0.998	0.357	0.513
PR	0.999	0.531	1.424	0.999	0.183	0.321

44 s, compared to RF's 67 s. Sample size can be an important factor when performing multiple parallel computations.

The optimized regressors were also analyzed in terms of model size (in KB), reflecting their complexity and storage requirements. The lightest models were PR (degree 12, 16 KB) and NN (3 hidden layers, 128 neurons, 438 KB). In contrast, RF (100 estimators, maximum depth of 40) required the largest storage, occupying 1.3 GB, which can be attributed to the large number of conditional ('if') operations involved in its estimation process.

To provide a graphical representation of the regressor estimations, a heatmap (Figure 3) was used to analyze the relationship between muscle activation and the predicted values of f_o and SPL. For this analysis, P_S was fixed at 1 kPa, which represent a mean value of the range. The results show that frequency increases with CT muscle activation, with a

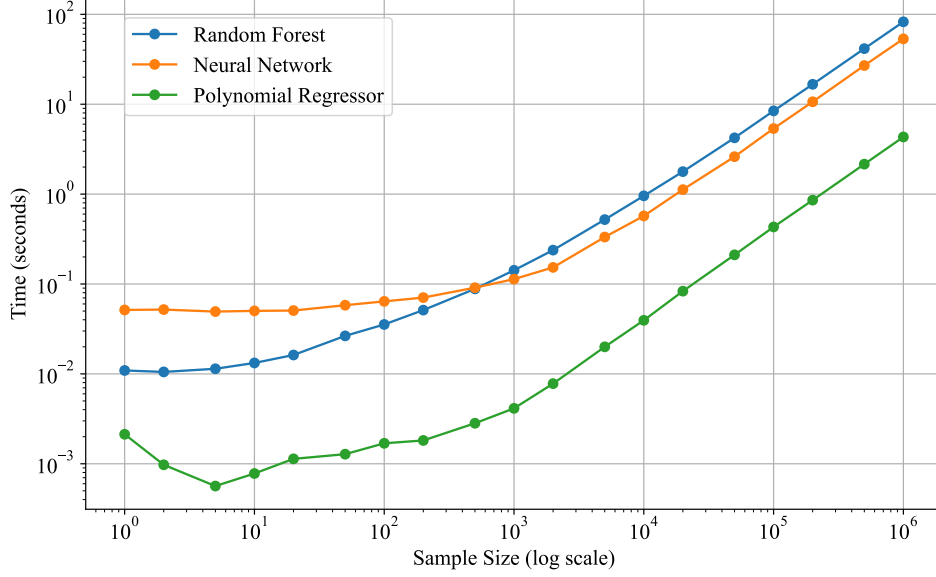


FIG. 2: Execution time vs sample size for the different ext-BCM-based regressors, using Random Forest (RF), Neural Network (NN) and Polynomial Regressor (PR).

more curved relationship at low frequencies and a linear relationship at higher frequencies. Notably, all regressors produced highly similar estimations, which can be attributed to the large number of training samples used. This aspect will be contrasted in the following section when evaluating regressors with the TBCM dataset.

On the other hand, regarding the capacity of the three regressors as a model of the plant in a trajectory tracking scheme, Figure 4 presents the tracking results for pitch glide (a) and loudness glide (b). The regressors effectively follow the reference trajectory, depicted as a black dashed line.

Regarding f_o and SPL references (i-ii), all regressors exhibit a smooth tracking behavior. However, a step-like pattern can be observed in the RF regressor, especially in the SPL trajectories, as it estimates values based on discrete samples from the training dataset. In

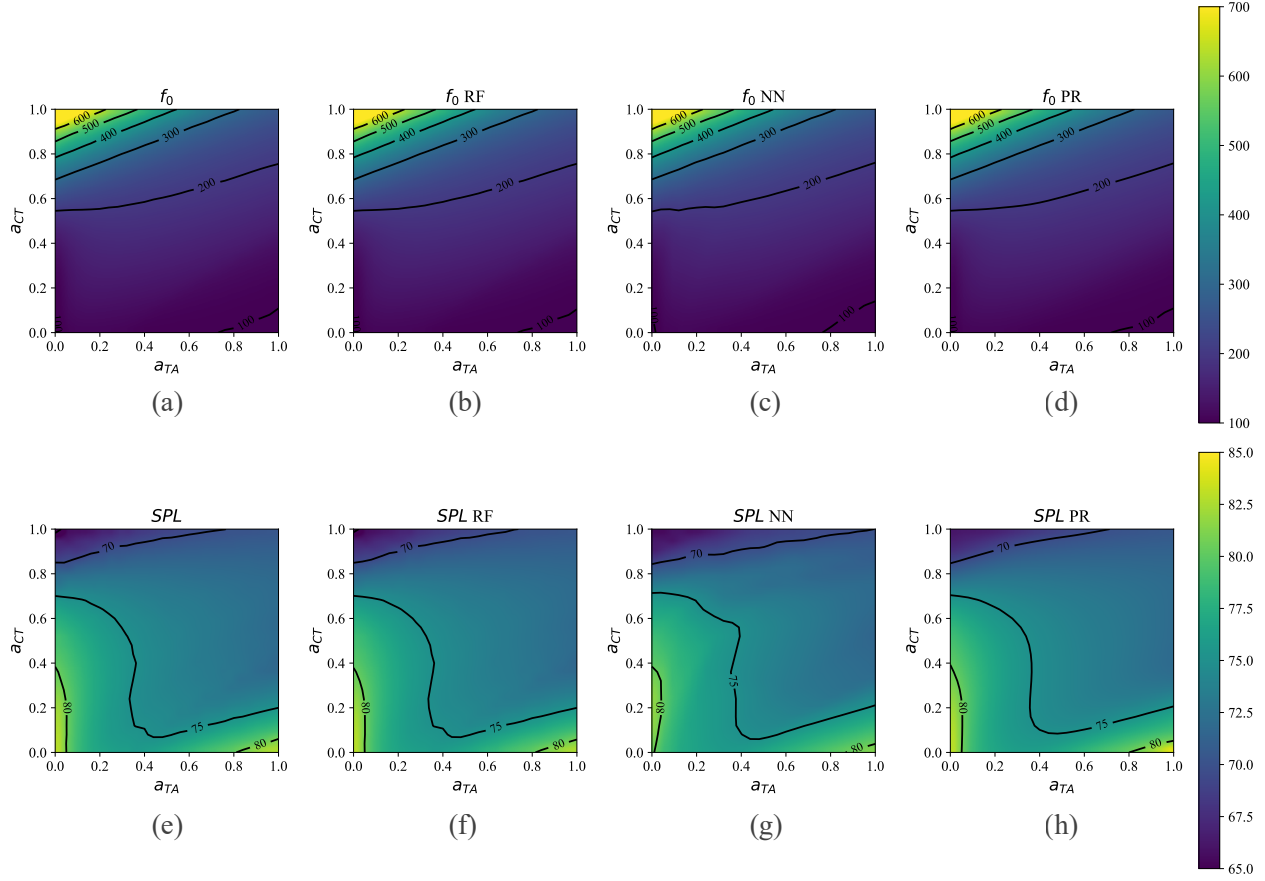


FIG. 3: MAP of f_o (a-d) in Hz and SPL (e-h) in dB for the different ext-BCM-based regressors. RF: Random forest. NN: Neural Network. PR: Polynomial regressor.

contrast, the NN and PR models generate continuous outputs. Despite these differences, all three regressors result in the same control input behavior for the plant (iii-v).

For pitch glide, an increase in CT activation and a slight decrease in TA activation are observed, aligning with their roles in the regulation of vocal fold tension (Chhetri and Park, 2016; Yin and Zhang, 2013). P_S tends to remain constant, as a fixed SPL was imposed.

In contrast, in the loudness glide, the CT muscle tends to remain constant while TA activation decreases to counterbalance the effect of P_S on f_o , leveraging its stabilizing role (Chhetri and Neubauer, 2015; Chung *et al.*, 2024). The increase in SPL, driven by P_S , aligns

with its strong correlation (Chhetri and Park, 2016; Lin *et al.*, 2020) and the fact that SPL is a logarithmic scale measured of sound pressure.

In general, ext-BCM-trained regressors successfully achieve smooth tracking of the proposed gestures, although with slightly less smoothness compared to the RF regressor.

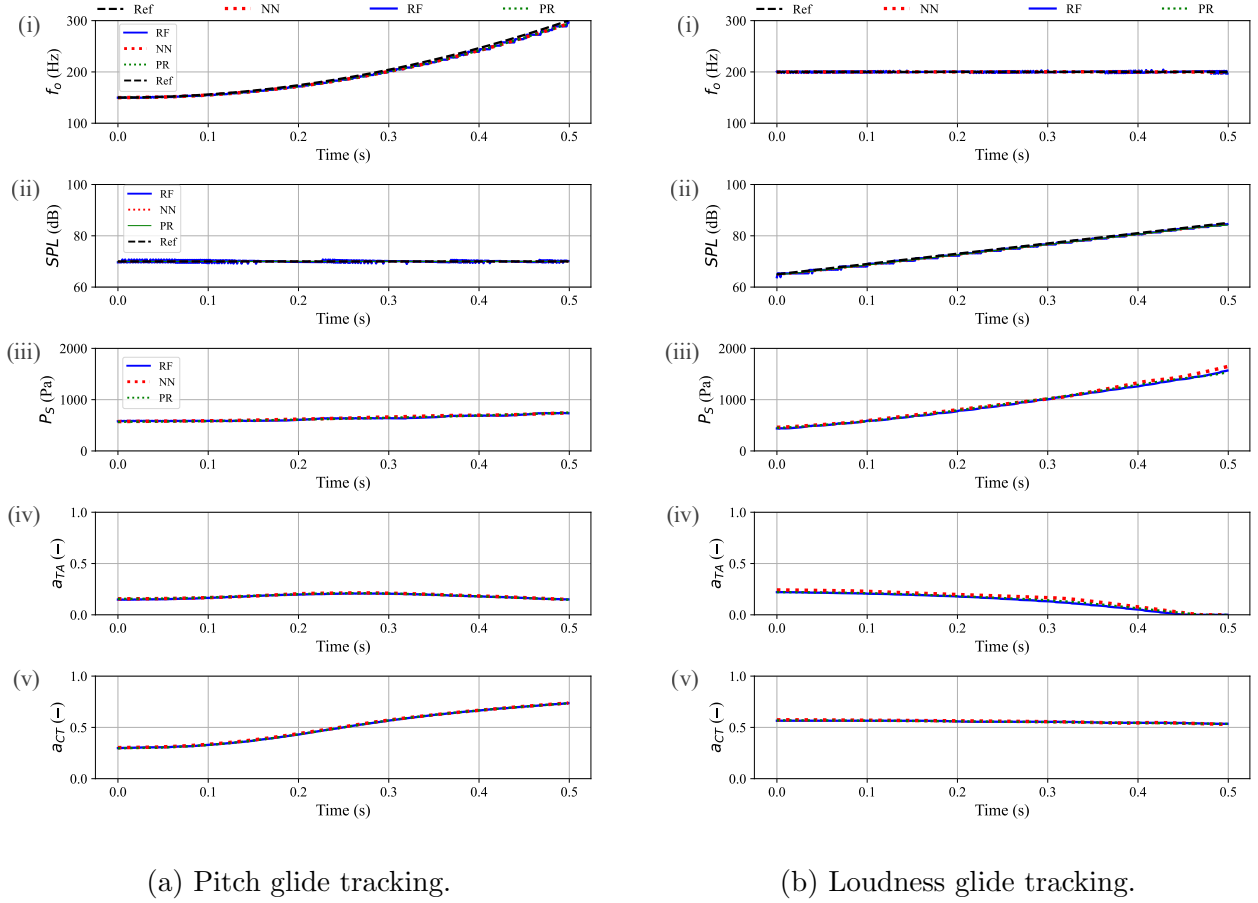


FIG. 4: Tracking glides using ext-BCM-based regressors. Target signals: (i) f_o and (ii) SPL. Input signals: (iii) subglottal pressure P_S , (iv) TA activation a_{TA} , and (v) CT activation a_{CT} . The black dashed line is the gesture target space.

B. Evaluation of machine learning regressors for forward mapping and reference tracking scheme using the TBCM Dataset

This analysis aims to evaluate how training with a more complex laryngeal representation, TBCM, which incorporates a more sophisticated muscle model (Titze and Hunter, 2007), affects performance, particularly under conditions of reduced dataset size. As stated previously, ML regressors also offer the possibility of adapting to new phonatory conditions. To explore their generalization to different contexts, we re-trained the regressors using the TBCM dataset while maintaining the optimized structure from the previous section. Table III presents the performance metrics, showing a noticeable increase in error compared to the ext-BCM dataset. However, the results remain within an acceptable range, with values of R^2 exceeding 0.94 in all cases. This increase in error is expected due to the significantly reduced number of training examples, with over one order of magnitude fewer data points. The trend previously observed persists, with RF exhibiting the lowest error (4 Hz for f_o and 1 dB for SPL). However, the performance gap has narrowed compared to the other regressors, which now exhibit errors of around 6 Hz for f_o and 2 dB for SPL.

Regarding the computational time for training, since the regressor structures remain unchanged, the execution times are the same as reported in Figure 2. However, a notable change was observed in the size of the RF model: Due to the smaller training dataset, its size was reduced to 140 MB, while the sizes of the other two regressors remained unchanged.

Heatmaps for f_o and SPL (Figures 5) illustrate the differences in phonatory representation between the datasets. The TBCM model introduces a more non-linear phonation

TABLE III: Results of optimal regressors for TBCM dataset, using Random Forest (RF), Neural Network (NN) and Polynomial Regressor (PR).

	f_o			SPL		
	R^2	MAE (Hz)	RMSE (Hz)	R^2	MAE (dB)	RMSE (dB)
RF	0.996	1.733	4.279	0.988	0.401	1.038
NN	0.993	3.668	5.676	0.948	1.346	2.203
PR	0.991	4.566	6.678	0.955	1.274	2.049

behavior, which affects the regressor predictions. A distinct pattern was observed in the relationship between f_o and muscle activation. The training data scale also allowed for differences in heatmap shapes between the regressors. SPL proved more challenging to reproduce, particularly for RF (Figure 5 f), where squared patterns emerged, likely aligned with its hierarchical structure. In contrast, PR (Figure 5 h), produced smoother patterns due to its polynomial nature, while NN, (Figure 5 g), generated SPL representations more consistent with the original dataset heatmap.

The tracking results for the TBCM-trained regressors are presented in Figure 6. While the regressors are capable of following the gesture trajectories, the RF model exhibits some instability.

In pitch glide (a), RF struggles to maintain stable SPL (ii), showing oscillatory behavior at certain time instances, with less problem in follows the increase in f_o (i). Regarding the

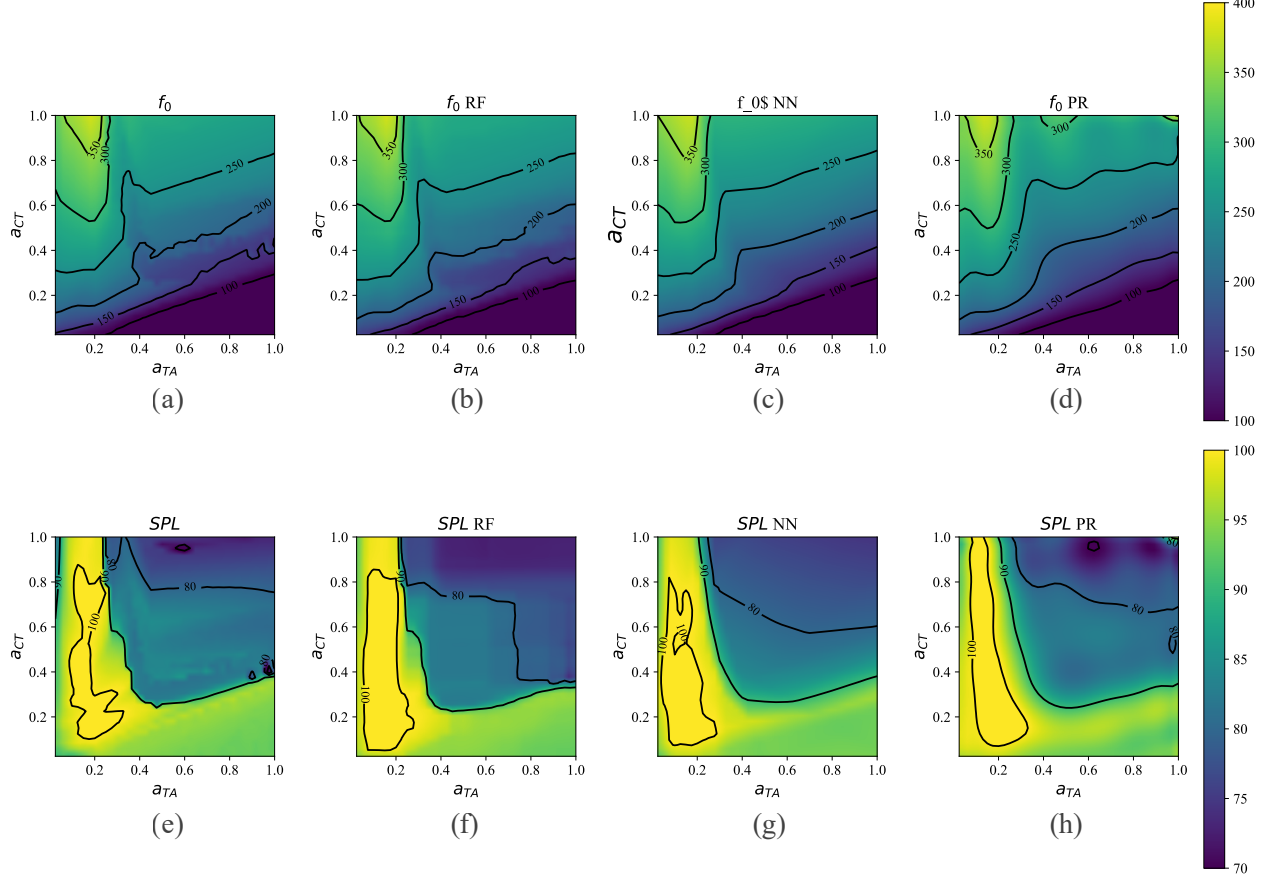


FIG. 5: MAP of f_o (a-d) and SPL (e-h) for the different TBCM-based regressors. RF:

Random forest. NN: Neural Network. PR: Polynomial regressor.

control inputs (iii-v), a similar pattern to that of ext-BCM is observed, where the f_o increase is primarily driven by CT activation. However, since the target frequency is 300 Hz, at the upper limit of the model range, discrepancies between the regressors appear at the end of the gesture: RF exhibits greater fluctuations, while NN increases P_S , which is consistence with increase in SPL. For the loudness glide (b), the increase in SPL is primarily driven by an increase in P_S . The P_S trajectory (iii) increases between the three regressors as time evolves, and muscle activations tend to remain constant, although differences between the regressors emerge toward the end of the gesture, with few increases in TA and decreases in

411 CT. RF again showing noticeable fluctuations. This behavior is expected, as 110 dB is at
 412 the upper limit of the range of the TBCM model.

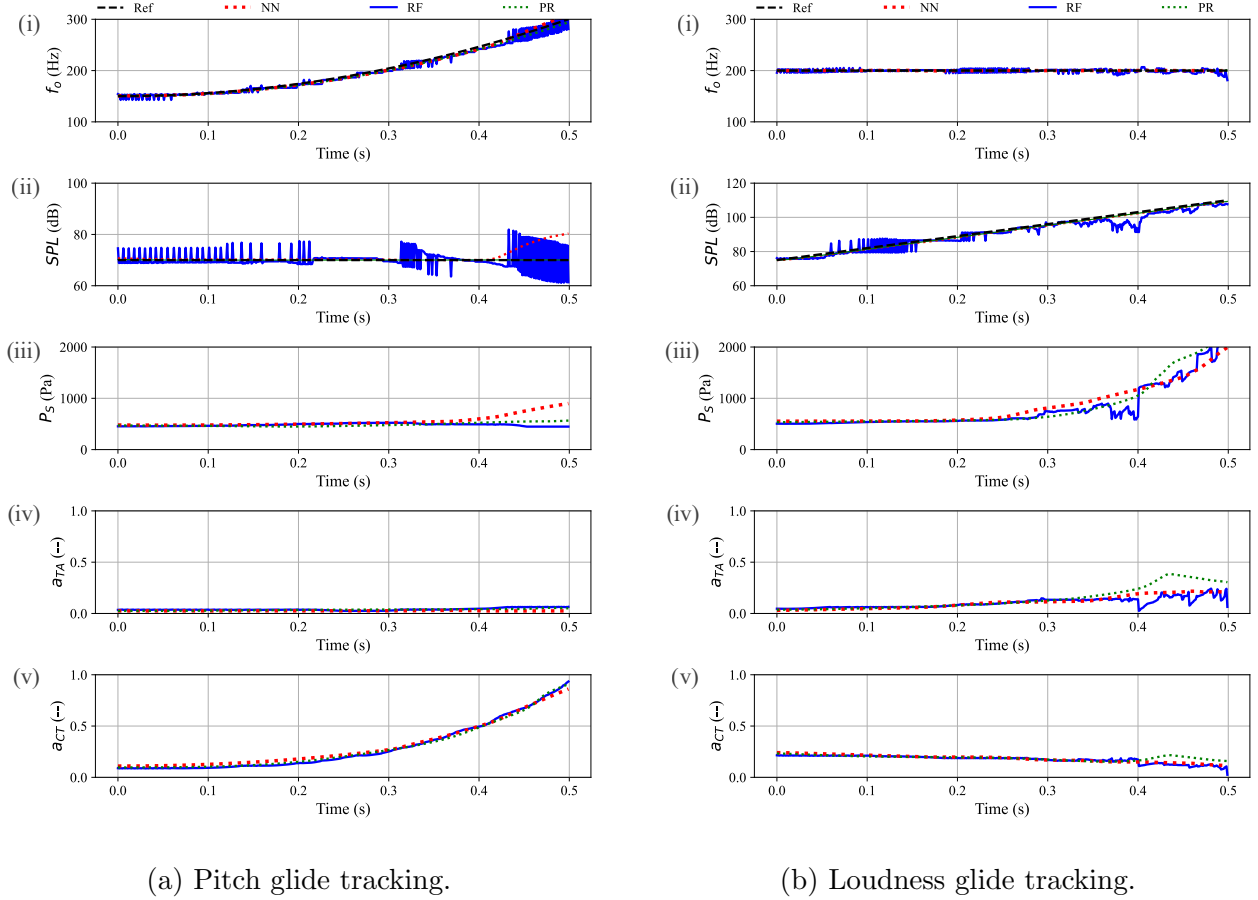


FIG. 6: Tracking glides using TBCM-based regressors. Target signals: (i) f_o and (ii) SPL. Input signals: (iii) subglottal pressure P_S , (iv) TA activation a_{TA} , and (v) CT activation a_{CT} . The black dashed line is the gesture target space, the green dotted line is the control phonation plant tracking, and the red dotted line is posterior gaped plant tracking.

IV. DISCUSSION

Tree-based models demonstrated outstanding adaptability to this type of data, achieving R^2 values of 0.999 and mean errors below 1 Hz for f_o and below 0.1 dB for SPL. In the ext-BCM condition, although all regressors reached high accuracy, NN and PR exhibited higher errors compared to RF. Specifically, NN showed a mean SPL error approximately 15 times greater than that of RF, while PR SPL error was about 8 times greater. For f_o , both NN and PR presented more than 3 times higher MAE compared to RF.

When using the TBCM dataset, which contained significantly fewer samples, these differences were reduced. The SPL MAE of NN was approximately 3.4 times higher than that of RF, while for PR it was approximately 3.2 times higher. In the case of f_o , NN doubled RF error, and PR error was about 2.6 times greater. These results indicate that as the size of the dataset decreases, the performance gap between regressors tends to diminish, likely because of the reduced capacity to exploit model complexity.

When considering execution time and regressor size, the efficiency of RF in terms of metrics contrasts with its memory requirements, as it needs significant space to structure the non-linear relationships between variables. In contrast, NN and PR only need to store matrix values, making them considerably smaller in size. Regarding execution time, PR requires only a simple matrix-vector multiplication, making it significantly faster than NN, which involves multiple such operations, or RF, which requires numerous conditional evaluations. Thus, PR is lightweight and fast compared to the other regressors.

The choice of ML regressors should be guided by the trade-off between predictive performance, model complexity, and computational resources (Choo and Chang, 2022). Therefore, error metrics alone should not be the only criterion in these applications. In control tasks, where multiple iterations of the plant are required, other factors must also be considered.

Regarding the ability to represent different laryngeal configurations, the results obtained with the TBCM model, which uses a smaller dataset, highlight a crucial aspect of such applications: the importance of dataset size. Differences in regressor applicability become evident as the number of data points decreases, a scenario that will inevitably arise when working with real-world data. This tracking analysis highlights the critical role of sample size in developing a robust plant for biomechanical control. Fine-grained adjustments are required to transform output discrepancies into accurate input corrections. A regressor that replicates training data well but lacks interpolation capability struggles with this aspect. Additionally, the lower number of training samples influences the polynomial representation of PR and the nonlinear behavior of NN, leading to potential deviations. In terms of retraining with a dataset, all three regressors have limitations. PR is trained analytically, meaning that changing the dataset while maintaining the polynomial order can introduce fitting errors. NN, on the other hand, would struggle to adjust its weight matrices with fewer samples, as avoiding overfitting would negatively impact performance. However, the most affected regressor is RF, which, as observed, increases its step-like behavior, preventing fine-grained estimation, an essential feature for the feedback component of motor control.

The integration of ML regressors not only enables the development of models tailored to individual subjects, incorporating specific anatomical features or pathologies, but also

facilitates the use of transfer learning techniques, addressing the issue of dataset size. This approach does not require training models from scratch; instead, it leverages prior training on a larger dataset. In the case of NN, these strategies are straightforward to implement and consist of fine-tuning selected layers using the subject-specific dataset (Ibarra *et al.*, 2025), an approach that has shown promising results. For RF, a similar strategy can be applied by constructing an ensemble of regressors trained on each dataset (Gu *et al.*, 2022), with predictions combined through weighted estimates of the two datasets, or by refining branch weights using subject-specific data once the baseline architecture has been defined (Segev *et al.*, 2016). For PR, closely related to linear models, two-stage transfer learning approaches have also been developed: the first stage involves training on the base dataset, followed by transfer through methods such as Lasso-type regularization (Li *et al.*, 2022) or fine-tuning with gradient descent iterations to minimize error on the subject-specific dataset (Obst *et al.*, 2022). These strategies demonstrate that such regressors provide practical tools to adapt to data-scarce contexts by exploiting an initial training phase on large, validated datasets.

To frame this within the context of biomechanical laryngeal control, a tracking experiment was conducted using an inverse Jacobian approach as feedback. The observed glide trends align with our understanding of laryngeal muscle activation and P_S . However, these glides did not capture variations in the TA muscle. This can be explained by two factors: (i) the existence of multiple solutions and the tendency of the inverse Jacobian approach to favor solutions with lower activation (minimizing the input vector), and (ii) the less direct role of TA compared to P_S and CT (Chhetri and Park, 2016; Chung *et al.*, 2024).

Future work offers several avenues. A natural extension of this study is the application of ML regressors to subject-specific cases, where individualized datasets could be used to improve the accuracy of existing transfer learning strategies. In the area of gesture tracking, the problem of non-unique inverse solutions may be addressed through constraints such as the null space formulation in the feedback loop (Chen and Walker, 1993). Additionally, one could explore the direct inclusion of ML regressors within the inverse feedback component (Palaparthi *et al.*, 2024), which aligns with previous efforts in laryngeal parameter estimation (Donhauser *et al.*, 2024; Ibarra *et al.*, 2025, 2021; Zhang, 2020).

V. CONCLUSION

The results of this study demonstrate that machine learning regression models are effective for forward mapping in laryngeal control, achieving high accuracy and low error in the prediction of fundamental frequency and sound pressure level. Among the evaluated methods, tree-based approaches, especially RF, performed best in capturing nonlinear input-output relationships, while PR and NN offer advantages in computational efficiency and model compactness.

Direct regression using machine learning techniques improves both modeling performance and adaptability to laryngeal variation, while also offering advantages in computational efficiency. This makes it a promising approach for customizing models to individuals with diverse vocal characteristics. Additionally, the sensitivity of model performance to dataset size underscores the importance of data availability in practical applications. Notably, as

the number of training samples decreases, differences among regressors tend to diminish, indicating a convergence in predictive behavior.

When large datasets are available and memory constraints are not a concern, RF provides the best trade-off between accuracy and robustness. Otherwise, a transfer learning approach using PR or NN may be more suitable. Transfer learning strategies, especially for NN, have been successfully applied in other speech-related tasks, demonstrating their potential in refining predictions with limited data. For RF and PR, alternative training methods, such as sequential or hybrid regressors, could mitigate performance degradation when dealing with smaller datasets.

Although preliminary, this finding marks an important starting point for future research, particularly given the growing interest in applying ML techniques to speech and vocal modeling. Future work will focus on implementing real examples of transfer learning with these regressors and extending ML applications to other components of the biomechanical control framework.

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AUTHOR DECLARATIONS

Conflict of interest

Dr. Matías Zañartu has financial interest in Lanek SPA, a company focused on developing and commercializing biomedical devices and technologies. Dr. Zañartu interests were reviewed and are managed by Universidad Técnica Federico Santa María in accordance with its conflict-of-interest policies.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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